Jyotish LoRA v4 Data Build: Progress Log

Oct 19, 2026 · @pawan

Zero-cost rebuild of the training data for the Vedic astrologer model: everything runs on the Claude Pro subscription, the local RTX 5070 Ti and code. This log is updated as steps finish; the PDF copy lives at `docs/v4-progress-log.pdf`.

Status at a glance

Stage 2A (data) is complete: 104,528 training rows, built 20 October 03:15. Stage 2B (evaluation) is complete except the owner's 60-item calibration ratings. Stage 2C (training) has started with a smoke test on the workstation GPU; the full run waits for the 200-row human review.

#	Step	Status	Result	Where
1	Python 3.11 environment, all tests passing	Done	180 tests pass	`C:\Users\mobirizer\.venvs\astro`
2	Corpus located and linked into the repo	Done	52 PDFs	`JioAICloud-Download/` junction
3	PDF extraction and chunking with OCR (RapidOCR, no admin needed)	Done	4,169 chunks, 45 books, 99.0% usable pages	`data/chunks/chunks.jsonl`
4	Fact tables v4 (Hinglish, contrast, yogas, vargas)	Done	737 rows	`data/sft_v4/facts_v4.jsonl`

#	Step	Status	Result	Where
5	Chart copying drills from 400 synthetic charts	Done	2,000 rows	`data/sft_v4/drills.jsonl`
6	Rule extraction from the 1,339 checked answers (Claude)	Done	8,833 rules, 80 min	`data/sft_v4/rules/`
7	Rule extraction from all chunks incl. OCR books (Claude)	Done	28,982 rules; merged base 37,029	`data/sft_v4/rules.jsonl`
8	Paraphrase generation: workstation GPU plus four Kaggle T4 kernels	Done	36,998 of 37,029 rules, finished 20 Oct 03:10	`runs/rule_qa_generate4.log`
9	Pair checks: containment, English-only with Hinglish rescue, safety block, citation, dedupe	Done	99,474 pairs kept (60%), 17,456 Hinglish	`data/sft_v4/rule_qa.jsonl`
10	Assembled v4 dataset (final)	Done	104,528 train rows, about 55M tokens	`data/sft_v4/train.jsonl`, `kaggle/astro-sft-v4/`

#	Step	Status	Result	Where
11	Evaluation v2: 267-item held-out test set with Opus references (Claude)	Done	267 items, 7 task types	`evalsets/test_v2.jsonl`
12	Evaluation v2: quiz (411 items), 100 adversarial safety prompts, chart-claim checker, two-judge runner, pre-registration	Done	ready; `astro-eval-v4` on both Kaggle accounts	`src/eval/`, `runs/preregistration_v4.md`
13	Human review of 200 rows (constitution IV.2)	Needs owner	CSV written, verdicts pending	`data/reviews/review_v4.csv`
14	Local training environment (CUDA torch, Unsloth) and scripts	Done	verified on the 5070 Ti, bf16	`C:\Users\mobirizer\.venvs\train`, `src/train/`
15	Training run v4: smoke test now; full run after the review	Running	smoke test running since 03:15	`runs/train_smoke.log`
16	Tesseract install (admin prompt)	Optional	RapidOCR used instead	

Running now

One job is running on the workstation GPU: the training smoke test (`src/train/train_local.py --smoke`), 30 steps of Qwen3-8B in 4-bit with Unsloth on 200 rows, to prove the training path before the full run. The paraphrase generation finished at 03:10 on 20 October.

Item	Value
Started	20 Oct 2026, 03:15 IST
What it checks	model and tokenizer load, LoRA attach (r=32, rsLoRA), completion-only loss, quiz callback, adapter save
Expected	10 to 20 minutes including the model download
Log	`runs/train_smoke.log`
Next	full run: 2 epochs over about 55M tokens, roughly 18 to 24 hours on this GPU, started after the 200-row review is imported

If the full run is too slow for the workstation, `build_v4` can cap the paraphrases at 2 pairs per rule (about 30M tokens), or the run can go to Kaggle with the two accounts' tokens, which are now on this machine.

Timeline

When (IST)	What happened
20 Oct 03:15	Final dataset built: 104,528 training rows; training smoke test started on the 5070 Ti
20 Oct 03:10	Paraphrase generation finished: 36,998 of 37,029 rules sampled
20 Oct 01:20	Training venv verified (CUDA torch, Unsloth, TRL, PEFT, bitsandbytes); training and local eval scripts written

When (IST)	What happened
20 Oct 00:40	Gate B code done: quiz v2 (411 items), chart-claim checker, two-judge runner, pre-registration; review CSV exported; `astro-eval-v4` uploaded to both Kaggle accounts
19 Oct 22:25	Kaggle outputs merged (7,634 rules); dataset rebuilt: 89,869 rows; workstation started on the last 5,488 rules
19 Oct 22:00	All four Kaggle kernels completed at the 11.3-hour cap; workstation finished shards 0 to 3
19 Oct 18:45	Evaluation v2 Claude jobs finished: 267 test items, 375 quiz items, 100 safety prompts
19 Oct 11:10	Evaluation v2 Claude jobs launched to use the subscription window
19 Oct 10:08	Four Kaggle kernels running on shards 4 to 7 after two startup fixes
19 Oct 09:30	Implementation plan rewritten as v2.0 (Phase 1 history and lessons, Phase 2 programme) with PDF
19 Oct 09:00	Safety filter added to the pair checker; first dataset: 29,022 rows
19 Oct 08:10	All rules merged: 37,029
19 Oct 02:10	OCR pass finished with RapidOCR: 4,169 chunks from 45 books
19 Oct 00:00	Answers rule extraction finished: 8,833 rules; paraphrase job launched
18 Oct 23:40	Books found; junction created; RapidOCR wired in
18 Oct 22:00	Fact tables and drills written and run
18 Oct 21:20	Review document and PDF of the full project

Dataset so far

The final v4 training set has 104,528 rows, 45 times v3, with validation (67) and test (60) unchanged so results stay comparable across versions. It is about 55M tokens (median 483 tokens per row), with 2,510 chart rows carrying a code-written reasoning block.

Stream	Rows	How it is made	How it is checked
v3 rows (teacher answers, v2 facts, strict safety)	2,327	unchanged from v3	as before
Fact tables v4	727	code, from the engine's reference tables; 10 phrasings per fact, Hinglish and contrast questions	correct by construction; quiz-overlap filter
Chart copying drills	2,000	code, from 400 fresh synthetic charts, v3 chart text, code-written working	correct by construction
Rule paraphrases	99,474	Claude extracted 37,029 atomic rules from answers and chunks; the local model and four Kaggle T4s rewrote each as 4 pairs	entity containment against the rule record, English-only answers with Hinglish rescue, safety block, citation ban, attribution present, question on rule, dedupe, no eval overlap

Rule base behind the paraphrases: 37,029 rules, 31,756 with accepted pairs (mean 3.1 per rule); 17,456 pairs carry Hinglish questions. Most attributed texts: Saravali, Brihat Parashara Hora Sastra, Crux of Astrology, Yavanajataka, Raman's How to Judge a Horoscope, Scientific Hindu Astrology, Vedic Astrology: An Integrated Approach, Garga Hora, Horasara, Phaladeepika.

Pair keep rate 60% of 164,975 generated; the main reject reasons are answers written in Hinglish against instructions (14,798 before rescue), an extra planet not in the rule (5,283), safety (death, disease, stigma), and off-rule questions. Evaluation v2 sets, all held out: 267 test items with Opus reference answers, 411 quiz items (36 code-graded, 375 judge-graded), 100 adversarial safety prompts. The chart-claim checker already shows the problem to fix: on Phase 1 chart answers, v2-8b's base had 96% of claims verified and the adapter 88%.

Next steps and decisions

In order. The first two happen automatically when the job ends; the rest need a word from you.

- [ ] Smoke test result (20 Oct morning); then the base-model baseline answers on the v2 sets
- [ ] **Your decision:** fill `data/reviews/review_v4.csv` (200 rows: approve, edit or reject). The full training run starts once it is imported, as the constitution requires
- [ ] Full v4 run on the 5070 Ti (about 18 to 24 hours for 2 epochs; or cap paraphrases at 2 per rule), then `eval_generate_local` and `run_eval_v2 judge` with Opus 5.5 and Sonnet 5.5
- [ ] **Your decision:** rate the 60-item judge calibration set
- [ ] Known-vs-unknown screen and general-replay rows, if the first v4 result calls for a data change
- [ ] Optional: approve the Tesseract installer's admin prompt if you prefer Tesseract over RapidOCR

How to check, resume and rebuild

All commands run from the repo root `C:\Users\mobirizer\Desktop\jyotish-lora` with the venv Python `C:\Users\mobirizer\.venvs\astro\Scripts\python.exe` (shown as `python` below).

Check progress of the running job:

```
type runs\rule_qa_generate2.log | findstr /R "rules"
```

Resume the job after a restart (it skips rules already sampled):

```
python -X utf8 -m src.dataset.rule_qa generate --k 4 --workers 4
```

Rebuild the dataset at any time from whatever has been generated:

```
python -X utf8 -m src.dataset.rule_qa build
python -X utf8 -m src.dataset.build_v4
```

Outputs: `data/sft_v4/train.jsonl`, `val.jsonl`, `test.jsonl`, `manifest.json`, and a Kaggle upload folder `kaggle/astro-sft-v4/`. Reports: `runs/rules_report.md`, `runs/rule_qa_report.md`, `runs/chunking_report.md`, `runs/extraction_report.md`.

Run the tests:

```
python -X utf8 -m pytest tests -q
```

The project's own handoff file, `STATUS.md` section 0, carries the same state and is updated whenever a step finishes.